

HBSC: Data Overview and Prep

Updated: 2026-08-14

```
rm(list = ls())

setwd("C:/MisLocalFiles/Github/HBSC/scripts")
project_folder <- "C:/MisLocalFiles/Github/HBSC/"

require(tidyverse)

## Loading required package: tidyverse

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.0      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

require(broom)

## Loading required package: broom

require(naniar)

## Loading required package: naniar

## Warning: package 'naniar' was built under R version 4.5.3

library(purrr)
library(patchwork)
library(knitr)
```

Dataset Overview

```
hbhc2018 <- read_delim(paste0(project_folder,
                               "/data/raw/HBSC20180Aed1.1.csv"),
                      delim = ";",
                      na = c("", " "))
```

```
## Rows: 244097 Columns: 120
## -- Column specification -----
## Delimiter: ";"
## chr   (5): region, weight, countryborn, countrybornmo, countrybornfa
## dbl (113): HBSC, seqno_int, cluster, countryno, id1, id2, id3, id4, adm, mon...
## num   (2): age, MBMI
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# working copy
dat <- hbsc2018
```

File has the following rows and columns:

```
dim(hbsc2018)
```

```
## [1] 244097    120
```

Pretty much all columns have missing values so need to address later. Though it seems much of the variables I will care about don't have much.

```
na_cnts <- data.frame(
  variable = names(dat),
  na_count = apply(dat, function(x) sum(is.na(x)))
)

row.names(na_cnts) <- NULL

na_cnts <- na_cnts %>%
  mutate(na_pct = round(100 * na_count / nrow(dat), 1)) %>%
  arrange(desc(na_count))

print(na_cnts)
```

```
##           variable na_count na_pct
## 1          agesex   228488   93.6
## 2   contraceptpill  228448   93.6
## 3   contraceptcondom 228070   93.4
## 4      employnotfa   224524   92.0
## 5      employnotmo   199890   81.9
## 6           hadsex   162986   66.8
## 7   cannabis30d_2   144442   59.2
## 8   cannabisltm_2   142745   58.5
## 9   countrybornfa    71901   29.5
## 10  countrybornmo    70659   28.9
## 11   countryborn    69118   28.3
## 12   talkstepmo     69006   28.3
## 13   talkstepfa     67025   27.5
## 14          grade    64830   26.6
## 15   oweight_who    54573   22.4
## 16          IOTF4    54541   22.3
```

## 17	MBMI	53311	21.8
## 18	bodyweight	41128	16.8
## 19	bodyheight	40036	16.4
## 20	emcsocmed8	32878	13.5
## 21	emcsocmed9	32855	13.5
## 22	emcsocmed7	32597	13.4
## 23	emcsocmed6	32544	13.3
## 24	emcsocmed5	32443	13.3
## 25	emcsocmed4	32428	13.3
## 26	emcsocmed3	32009	13.1
## 27	emcsocmed1	31973	13.1
## 28	emcsocmed2	31659	13.0
## 29	emconlpref3	31097	12.7
## 30	emconlpref2	30586	12.5
## 31	emconlpref1	29813	12.2
## 32	emconlfreq2	29552	12.1
## 33	emconlfreq4	29003	11.9
## 34	emconlfreq3	28973	11.9
## 35	emconlfreq1	26720	10.9
## 36	employmo	24135	9.9
## 37	employfa	23999	9.8
## 38	bulliedothers	21579	8.8
## 39	id3	20807	8.5
## 40	smok30d_2	20687	8.5
## 41	smokltm	20451	8.4
## 42	drunk30d	20415	8.4
## 43	famdec	20237	8.3
## 44	famsup	20091	8.2
## 45	famtalk	20019	8.2
## 46	alc30d_2	19326	7.9
## 47	drunkltm	19266	7.9
## 48	famhelp	18896	7.7
## 49	teachercare	18340	7.5
## 50	alcltm	18007	7.4
## 51	teachertrust	17839	7.3
## 52	talkfather	17491	7.2
## 53	breakfastwe	17468	7.2
## 54	studaccept	17444	7.1
## 55	beenbullied	16962	6.9
## 56	teacheraccept	16764	6.9
## 57	studhelpful	16746	6.9
## 58	studtogether	16351	6.7
## 59	talkmother	15521	6.4
## 60	cbeenbullied	15084	6.2
## 61	cbulliedothers	14934	6.1
## 62	fmeal	14738	6.0
## 63	IRFAS	12599	5.2
## 64	IRRELFAS_LMH	12599	5.2
## 65	fasdishwash	12124	5.0
## 66	thinkbody	11657	4.8
## 67	elsehome1_2	11124	4.6
## 68	fosterhome1	10938	4.5
## 69	sleepdifficulty	10814	4.4
## 70	friendshare	10813	4.4

## 71	friendtalk	10808	4.4
## 72	breakfastwd	10748	4.4
## 73	fasbathroom	10743	4.4
## 74	stepfahome1	10451	4.3
## 75	stepmohome1	10449	4.3
## 76	friendcounton	10374	4.2
## 77	motherhome1	10235	4.2
## 78	fatherhome1	10163	4.2
## 79	friendhelp	9744	4.0
## 80	injured12m	8974	3.7
## 81	fight12m	8762	3.6
## 82	dizzy	8566	3.5
## 83	toothbr	7910	3.2
## 84	fasholidays	7727	3.2
## 85	feellow	6696	2.7
## 86	fasbedroom	6672	2.7
## 87	backache	6543	2.7
## 88	irritable	6539	2.7
## 89	nervous	6525	2.7
## 90	physact60	6215	2.5
## 91	fasfamcar	6195	2.5
## 92	vegetables_2	6021	2.5
## 93	sweets_2	6004	2.5
## 94	softdrinks_2	5906	2.4
## 95	stomachache	5679	2.3
## 96	fascomputers	5527	2.3
## 97	schoolpressure	5486	2.2
## 98	fruits_2	5294	2.2
## 99	likeschool	5109	2.1
## 100	timeexe	4799	2.0
## 101	lifesat	4512	1.8
## 102	headache	4505	1.8
## 103	health	2577	1.1
## 104	age	1516	0.6
## 105	agecat	1516	0.6
## 106	yearbirth	1516	0.6
## 107	monthbirth	1106	0.5
## 108	month	18	0.0
## 109	HBSC	0	0.0
## 110	seqno_int	0	0.0
## 111	cluster	0	0.0
## 112	countryno	0	0.0
## 113	region	0	0.0
## 114	id1	0	0.0
## 115	id2	0	0.0
## 116	id4	0	0.0
## 117	weight	0	0.0
## 118	adm	0	0.0
## 119	year	0	0.0
## 120	sex	0	0.0

seqno_int

This is the unique identifier of each record and I confirmed it's unique.

```
length(unique(dat$seqno_int)) == nrow(dat)
```

```
## [1] TRUE
```

countryno

Number of countries in dataset is:

```
length(unique(dat$countryno))
```

```
## [1] 47
```

And this is the count of survey responses by country:

```
recode_map <- c(
  "8000"   = "Albania",
  "31000"  = "Azerbaijan",
  "40000"  = "Austria",
  "51000"  = "Armenia",
  "56001"  = "Belgium (Flemish)",
  "56002"  = "Belgium (French)",
  "100000" = "Bulgaria",
  "124000" = "Canada",
  "191000" = "Croatia",
  "203000" = "Czech Republic",
  "208000" = "Denmark",
  "233000" = "Estonia",
  "246000" = "Finland",
  "250000" = "France",
  "268000" = "Georgia",
  "276000" = "Germany",
  "300000" = "Greece",
  "304000" = "Greenland",
  "348000" = "Hungary",
  "352000" = "Iceland",
  "372000" = "Ireland",
  "376000" = "Israel",
  "380000" = "Italy",
  "398000" = "Kazakhstan",
  "428000" = "Latvia",
  "440000" = "Lithuania",
  "442000" = "Luxembourg",
  "470000" = "Malta",
  "498000" = "Republic of Moldova",
  "528000" = "Netherlands",
  "578000" = "Norway",
  "616000" = "Poland",
  "620000" = "Portugal",
```

```

"642000" = "Romania",
"643000" = "Russia",
"688000" = "Serbia",
"703000" = "Slovakia",
"705000" = "Slovenia",
"724000" = "Spain",
"752000" = "Sweden",
"756000" = "Switzerland",
"792000" = "Turkey",
"804000" = "Ukraine",
"807000" = "Macedonia",
"826001" = "England",
"826002" = "Scotland",
"826003" = "Wales",
"826004" = "Northern Ireland",
"840000" = "USA"
)

dat$country_name <- recode_map[as.character(dat$countryno)]

dat %>%
  group_by(country_name) %>%
  summarise(n = n(),
            .groups = "drop") %>%
  mutate(pct = round(100 * n / sum(n),1)) %>%
  arrange(desc(n)) %>%
  print(n=Inf)

```

```

## # A tibble: 46 x 3
##   country_name      n  pct
##   <chr>          <int> <dbl>
## 1 Wales          15951  6.5
## 2 Canada         12950  5.3
## 3 Czech Republic 11564  4.7
## 4 France          9170  3.8
## 5 <NA>            8411  3.4
## 6 Israel          7712  3.2
## 7 Switzerland     7510  3.1
## 8 Iceland         6996  2.9
## 9 Ukraine         6660  2.7
## 10 Portugal        6126  2.5
## 11 Turkey          5848  2.4
## 12 Slovenia        5667  2.3
## 13 Belgium (French) 5578  2.3
## 14 Poland          5224  2.1
## 15 Croatia         5169  2.1
## 16 Scotland        5021  2.1
## 17 Kazakhstan      4868  2
## 18 Slovakia        4785  2
## 19 Estonia         4725  1.9
## 20 Armenia         4717  1.9
## 21 Netherlands     4698  1.9
## 22 Republic of Moldova 4686  1.9

```

## 23 Macedonia	4658	1.9
## 24 Azerbaijan	4586	1.9
## 25 Romania	4567	1.9
## 26 Latvia	4412	1.8
## 27 Germany	4347	1.8
## 28 Belgium (Flemish)	4333	1.8
## 29 Spain	4320	1.8
## 30 Russia	4281	1.8
## 31 Georgia	4242	1.7
## 32 Sweden	4185	1.7
## 33 Italy	4144	1.7
## 34 Austria	4129	1.7
## 35 Luxembourg	4070	1.7
## 36 Serbia	3933	1.6
## 37 Ireland	3833	1.6
## 38 Lithuania	3797	1.6
## 39 Hungary	3789	1.6
## 40 England	3397	1.4
## 41 Denmark	3181	1.3
## 42 Finland	3146	1.3
## 43 Norway	3127	1.3
## 44 Malta	2576	1.1
## 45 Albania	1765	0.7
## 46 Greenland	1243	0.5

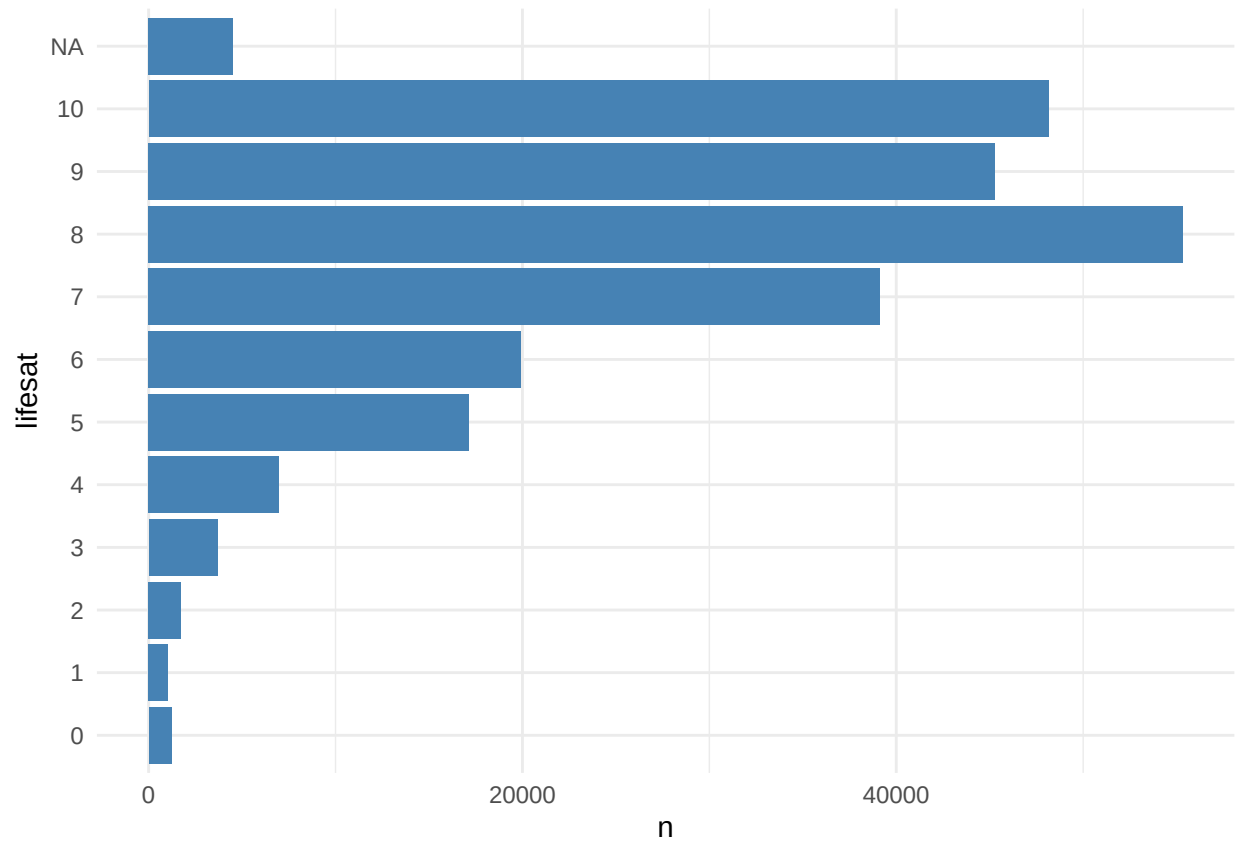
Variables to Predict

lifesat

Q: Here is a picture of a ladder. The top of the ladder “10” is the best possible life for you and the bottom “0” is the worst possible life for you. In general, where on the ladder do you feel you stand at the moment?

A: 0-10

```
ggplot(dat, aes(y = factor(lifesat))) +
  geom_bar(fill = "steelblue") +
  labs(
    y = "lifesat",
    x = "n"
  ) +
  theme_minimal()
```



lifesat is dichotomized for modeling as lifesat_low:

```
dat <- dat %>%
  mutate(
    lifesat_low = case_when(
      between(lifesat, 0, 5) ~ 1,
      between(lifesat, 6, 10) ~ 0,
      TRUE ~ NA))

table(dat$lifesat,
      dat$lifesat_low,
      useNA = "always"
    )
```

```
##
##      0      1  <NA>
##  0      0 1218      0
##  1      0 1041      0
##  2      0 1748      0
##  3      0 3691      0
##  4      0 6961      0
##  5      0 17137     0
##  6 19921      0      0
##  7 39120      0      0
##  8 55325      0      0
##  9 45271      0      0
## 10 48152      0      0
```



```
##    <NA>      0      0 4512
```

Ultimately the distribution is:

```
dat %>%  
  group_by(lifesat_low) %>%  
  summarise(n = n()) %>%  
  mutate(pct = n / sum(n))
```

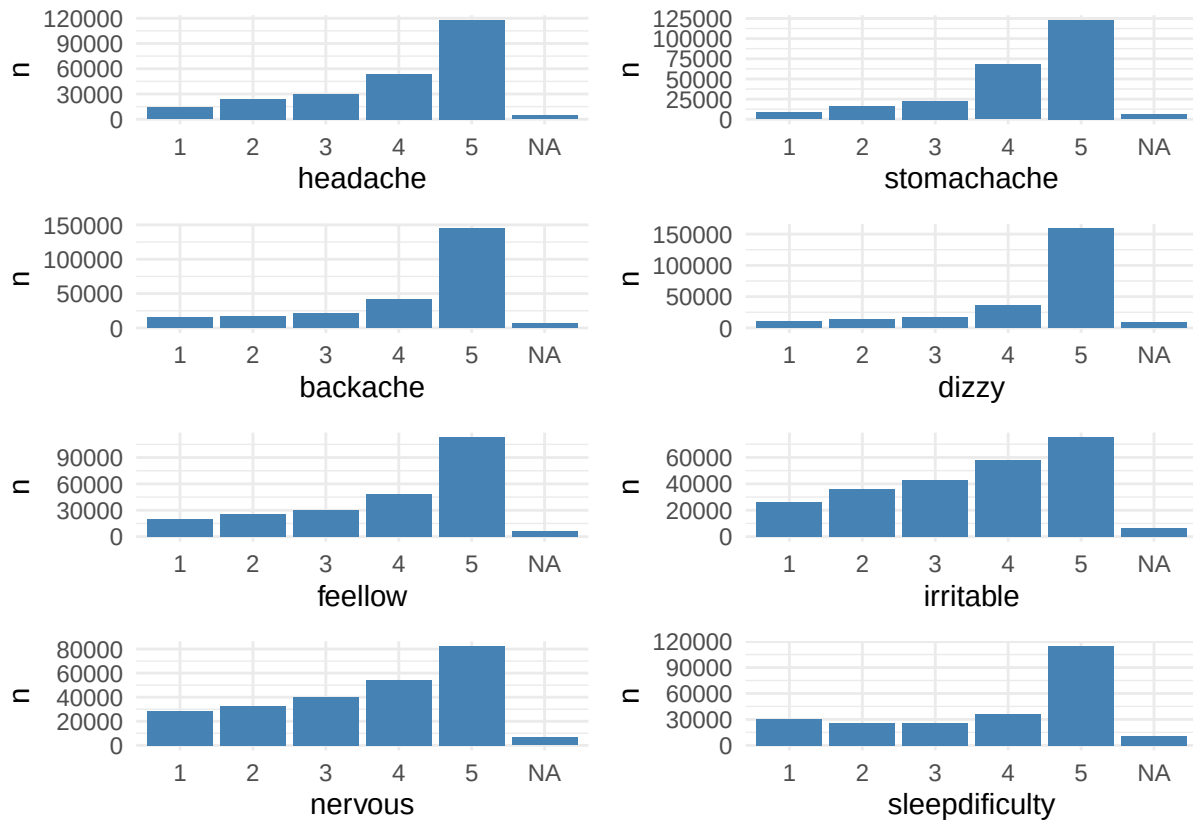
```
## # A tibble: 3 x 3  
##   lifesat_low      n    pct  
##       <dbl> <int> <dbl>  
## 1         0 207789 0.851  
## 2         1  31796 0.130  
## 3        NA   4512 0.0185
```

(mental) health complaints

There are 8 variables for this that reply to a question like: In the last 6 months: how often have you had the following...? And the answers goes from 1 About every day to 5 Rarely or never. The 8 variables are: headache, stomachache, backache, dizzy, feellow, irritable, nervous, sleepdifficulty.

Their distributions look like this:

```
vars_health_complaints <- c("headache", "stomachache", "backache", "dizzy",  
                             "feellow", "irritable", "nervous", "sleepdifficulty")  
  
plot_list <- map(vars_health_complaints, function(v) {  
  ggplot(dat, aes(y = factor(.data[[v]]))) +  
    geom_bar(fill = "steelblue") +  
    coord_flip() +  
    labs(  
      y = v,  
      x = "n"  
    ) +  
    theme_minimal()  
})  
  
wrap_plots(plot_list, ncol = 2)
```



These variables are then recoded to a binary where it flags if the complaint occurs more than once per week (1 or 2).

```
dat <- dat %>%
  mutate(
    headache_frequent = ifelse(headache <= 2, 1, 0),
    stomachache_frequent = ifelse(stomachache <= 2, 1, 0),
    backache_frequent = ifelse(backache <= 2, 1, 0),
    dizzy_frequent = ifelse(dizzy <= 2, 1, 0),
    feellow_frequent = ifelse(feellow <= 2, 1, 0),
    irritable_frequent = ifelse(irritable <= 2, 1, 0),
    nervous_frequent = ifelse(nervous <= 2, 1, 0),
    sleepdifficulty_frequent = ifelse(sleepdifficulty <= 2, 1, 0)
  )
```

The recoded variables look like this:

```
vars_health_complaints_frequent <- paste0(vars_health_complaints,
                                           "_frequent")

for (i in seq_along(vars_health_complaints)) {
  cat("\n\n###", vars_health_complaints[i], "x", vars_health_complaints_frequent[i], "\n\n")
  print(table(dat[[vars_health_complaints[i]]], dat[[vars_health_complaints_frequent[i]]], useNA = "always"))
}
```

##

```

##
## ### headache x headache_frequent
##
##          0      1   <NA>
## 1          0 13883      0
## 2          0 24257      0
## 3       30076      0      0
## 4       53676      0      0
## 5      117700      0      0
## <NA>        0      0  4505
##
##
## ### stomachache x stomachache_frequent
##
##          0      1   <NA>
## 1          0   8415      0
## 2          0 16223      0
## 3       22895      0      0
## 4       68018      0      0
## 5      122867      0      0
## <NA>        0      0  5679
##
##
## ### backache x backache_frequent
##
##          0      1   <NA>
## 1          0 14872      0
## 2          0 16137      0
## 3       21410      0      0
## 4       40910      0      0
## 5      144225      0      0
## <NA>        0      0  6543
##
##
## ### dizzy x dizzy_frequent
##
##          0      1   <NA>
## 1          0 10461      0
## 2          0 14272      0
## 3       17261      0      0
## 4       35189      0      0
## 5      158348      0      0
## <NA>        0      0  8566
##
##
## ### feellow x feellow_frequent
##
##          0      1   <NA>
## 1          0 20224      0
## 2          0 25739      0
## 3       29784      0      0
## 4       48898      0      0
## 5      112756      0      0
## <NA>        0      0  6696

```

```
##
##
## ### irritable x irritable_frequent
##
##           0      1  <NA>
## 1           0 26053      0
## 2           0 35879      0
## 3      42648      0      0
## 4      57922      0      0
## 5      75056      0      0
## <NA>      0      0 6539
##
##
## ### nervous x nervous_frequent
##
##           0      1  <NA>
## 1           0 28306      0
## 2           0 32770      0
## 3      40064      0      0
## 4      54109      0      0
## 5      82323      0      0
## <NA>      0      0 6525
##
##
## ### sleepdifficulty x sleepdifficulty_frequent
##
##           0      1  <NA>
## 1           0 30501      0
## 2           0 25543      0
## 3      26053      0      0
## 4      36570      0      0
## 5     114616      0      0
## <NA>      0      0 10814
```

The frequency of complaints is then added to calculate one sum for physical and one for mental. If the sum is 2 or more then it is determined that there is a health issue. Most papers do it for psychosomatic (all 8 vars) but I will focus on the 4 mental ones. Still, I include both vars in case.

```
dat <- dat %>%
  # add the frequency of complaints
  mutate(mental_complaints_sum = feellow_frequent +
        irritable_frequent +
        nervous_frequent +
        sleepdifficulty_frequent,
        physical_complaints_sum = headache_frequent +
        stomachache_frequent +
        backache_frequent +
        dizzy_frequent
  ) %>%
  # derive index based on sum
  mutate(mental_complaintsY = ifelse(mental_complaints_sum >=2, 1, 0),
        health_complaintsY = ifelse((mental_complaints_sum + physical_complaints_sum) >=2, 1, 0)
  )
```

```
cat("\n\n### mental_complaintsY x mental_complaints_sum\n")
```

```
##
##
## ### mental_complaintsY x mental_complaints_sum
```

```
table(dat$mental_complaints_sum,
      dat$mental_complaintsY,
      useNA = "always")
```

```
##
##           0      1   <NA>
##  0    121573      0      0
##  1     45509      0      0
##  2         0  29991      0
##  3         0  21172      0
##  4         0  12970      0
##  <NA>      0      0  12882
```

```
cat("\n\n### (health_complaintsY x health_complaints_sum\n")
```

```
##
##
## ### (health_complaintsY x health_complaints_sum
```

```
table(dat$mental_complaints_sum + dat$physical_complaints_sum,
      dat$health_complaintsY,
      useNA = "always")
```

```
##
##           0      1   <NA>
##  0    103678      0      0
##  1     41923      0      0
##  2         0  29221      0
##  3         0  21360      0
##  4         0  14640      0
##  5         0   8824      0
##  6         0   5202      0
##  7         0   2839      0
##  8         0   2219      0
##  <NA>      0      0  14191
```

mental_complaintsY

Ultimately our variable to predict will be mental_complaintsY; at ~26%.

```
dat %>%
  group_by(mental_complaintsY) %>%
  summarise(n = n()) %>%
  mutate(pct = n / sum(n))
```

```
## # A tibble: 3 x 3
##   mental_complaintsY      n    pct
##             <dbl> <int> <dbl>
## 1               0 167082 0.684
## 2               1  64133 0.263
## 3              NA  12882 0.0528
```

The Predictors

Problematic social media use (PMSU)

PMSU are the main variables of interest. There are 9 of them: emcsocmed1 to emcsocmed9. The prequestion for all of them is: We are interested in your experiences with social media. The term social media refers to social network sites (e.g. Facebook, [add other local examples]) and instant messengers (e.g. [insert local examples], WhatsApp, Snapchat, Facebook messenger). During the past year, have you... Please tick one box for each line.

The possible answers are 1 (no) and 2 (yes). Also those with 99 mean Missing due to skip pattern.

The topic of each question is:

```
questions <- data.frame(
  item = c(
    "emcsocmed1", "emcsocmed2", "emcsocmed3", "emcsocmed4",
    "emcsocmed5", "emcsocmed6", "emcsocmed7", "emcsocmed8", "emcsocmed9"
  ),
  topic = c(
    "Social media: I can't think of anything else",
    "Social media: Dissatisfied cause want to spend more time",
    "Social media: Felt bad cause want to play",
    "Social media: Failed to spend less time",
    "Social media: Neglected other activities",
    "Social media: Had arguments because of use",
    "Social media: Lied about amount of time spent",
    "Social media: Use it to escape from negative feelings",
    "Social media: Conflict with family because of use"
  )
)

kable(questions) #, caption = "Social Media Use Items")
```

item	topic
emcsocmed1	Social media: I can't think of anything else
emcsocmed2	Social media: Dissatisfied cause want to spend more time
emcsocmed3	Social media: Felt bad cause want to play
emcsocmed4	Social media: Failed to spend less time
emcsocmed5	Social media: Neglected other activities
emcsocmed6	Social media: Had arguments because of use
emcsocmed7	Social media: Lied about amount of time spent
emcsocmed8	Social media: Use it to escape from negative feelings
emcsocmed9	Social media: Conflict with family because of use

Since I need to get a sum of them I will recode them from 1/2 to 0/1. These are the distributions of the answers. I put the 99 as NA - is that correct?

The recoding looks like this:

```
# recode emcsocmed to 0/1 from 1/2
recode_emcsocmed <- function(old_col){
  new_col <- case_when(
    old_col == 1 ~ 0,
    old_col == 2 ~ 1,
    TRUE ~ NA #99?
  )
  return(new_col)
}

dat <- dat %>%
  mutate(emcsocmed1_r = recode_emcsocmed(emcsocmed1),
         emcsocmed2_r = recode_emcsocmed(emcsocmed2),
         emcsocmed3_r = recode_emcsocmed(emcsocmed3),
         emcsocmed4_r = recode_emcsocmed(emcsocmed4),
         emcsocmed5_r = recode_emcsocmed(emcsocmed5),
         emcsocmed6_r = recode_emcsocmed(emcsocmed6),
         emcsocmed7_r = recode_emcsocmed(emcsocmed7),
         emcsocmed8_r = recode_emcsocmed(emcsocmed8),
         emcsocmed9_r = recode_emcsocmed(emcsocmed9)
  )

vars <- paste0("emcsocmed", 1:9)

for (v in vars) {
  cat("## Table for", v)
  print(table(dat[[v]], dat[[paste0(v, "_r")]], useNA = "always"))
  cat("\n\n")
}
```

```
## ## Table for emcsocmed1
##           0      1  <NA>
##  1    156598      0      0
##  2         0  44267      0
##  99         0      0  11259
## <NA>         0      0  31973
##
##
## ## Table for emcsocmed2
##           0      1  <NA>
##  1    163166      0      0
##  2         0  38013      0
##  99         0      0  11259
## <NA>         0      0  31659
##
##
## ## Table for emcsocmed3
##           0      1  <NA>
##  1    157891      0      0
```

```

##      2      0 42938      0
##     99      0      0 11259
##    <NA>      0      0 32009
##
##
## ## Table for emcsocmed4
##           0      1  <NA>
##    1  138631      0      0
##    2      0 61779      0
##   99      0      0 11259
##  <NA>      0      0 32428
##
##
## ## Table for emcsocmed5
##           0      1  <NA>
##    1  168967      0      0
##    2      0 31428      0
##   99      0      0 11259
##  <NA>      0      0 32443
##
##
## ## Table for emcsocmed6
##           0      1  <NA>
##    1  162482      0      0
##    2      0 37812      0
##   99      0      0 11259
##  <NA>      0      0 32544
##
##
## ## Table for emcsocmed7
##           0      1  <NA>
##    1  171121      0      0
##    2      0 29120      0
##   99      0      0 11259
##  <NA>      0      0 32597
##
##
## ## Table for emcsocmed8
##           0      1  <NA>
##    1  137767      0      0
##    2      0 62193      0
##   99      0      0 11259
##  <NA>      0      0 32878
##
##
## ## Table for emcsocmed9
##           0      1  <NA>
##    1  171010      0      0
##    2      0 28973      0
##   99      0      0 11259
##  <NA>      0      0 32855

```

And this gives an idea of how correlated they are with each other.

Is this right? Too correlated? Or it's done different? It has to be done another way?


```
vars <- paste0("emcsocmed", 1:9)
```

```
r <- cor(dat[,vars],
        use = "complete.obs")
```

```
round(r,4)
```

```
##          emcsocmed1 emcsocmed2 emcsocmed3 emcsocmed4 emcsocmed5 emcsocmed6
## emcsocmed1      1.0000      0.9998      0.9998      0.9997      0.9998      0.9998
## emcsocmed2      0.9998      1.0000      0.9998      0.9997      0.9998      0.9998
## emcsocmed3      0.9998      0.9998      1.0000      0.9997      0.9998      0.9998
## emcsocmed4      0.9997      0.9997      0.9997      1.0000      0.9997      0.9997
## emcsocmed5      0.9998      0.9998      0.9998      0.9997      1.0000      0.9998
## emcsocmed6      0.9998      0.9998      0.9998      0.9997      0.9998      1.0000
## emcsocmed7      0.9998      0.9998      0.9998      0.9998      0.9998      0.9998
## emcsocmed8      0.9997      0.9997      0.9997      0.9997      0.9997      0.9997
## emcsocmed9      0.9998      0.9998      0.9998      0.9998      0.9998      0.9998
##          emcsocmed7 emcsocmed8 emcsocmed9
## emcsocmed1      0.9998      0.9997      0.9998
## emcsocmed2      0.9998      0.9997      0.9998
## emcsocmed3      0.9998      0.9997      0.9998
## emcsocmed4      0.9998      0.9997      0.9998
## emcsocmed5      0.9998      0.9997      0.9998
## emcsocmed6      0.9998      0.9997      0.9998
## emcsocmed7      1.0000      0.9998      0.9999
## emcsocmed8      0.9998      1.0000      0.9998
## emcsocmed9      0.9999      0.9998      1.0000
```

PMSU variables are then combined so output is represented in one variable, instead of 9. I will do 2 versions:
a) Yes/No where Y(6-9) and N(0-5) b) H/M/L where H(6-9), M(2-5) and L(0-1)

The distribution of PMSU categorized is:

```
vars_emcsocmed <- paste0("emcsocmed", 1:9)
vars_emcsocmed_r <- paste0("emcsocmed", 1:9, "_r")

dat <- dat %>%
  mutate(emcsocmed_sum = rowSums(dat[,vars_emcsocmed_r])) %>%
  mutate(
    pmsu_yn = case_when(
      between(emcsocmed_sum,6,9) ~ 1,
      between(emcsocmed_sum,0,5) ~ 0,
      TRUE ~ NA
    ),
    pmsu_lmh = case_when(
      between(emcsocmed_sum,6,9) ~ "High",
      between(emcsocmed_sum,2,5) ~ "Med",
      between(emcsocmed_sum,0,1) ~ "Low",
      TRUE ~ NA
    )
  )

cat("## Table for emcsocmed_sum x pmsu_yn")
```

```
## ## Table for emcsocmed_sum x pmsu_yn
```

```
table(dat$emcsocmed_sum,
      dat$pmsu_yn,
      useNA = "always")
```

```
##
##           0      1  <NA>
##  0      72474      0      0
##  1      36277      0      0
##  2      26066      0      0
##  3      19012      0      0
##  4      14908      0      0
##  5      10306      0      0
##  6           0  5905      0
##  7           0  3382      0
##  8           0  1878      0
##  9           0  2979      0
## <NA>         0      0 50910
```

```
cat("## Table for emcsocmed_sum x pmsu_lmh")
```

```
## ## Table for emcsocmed_sum x pmsu_lmh
```

```
table(dat$emcsocmed_sum,
      dat$pmsu_lmh,
      useNA = "always")
```

```
##
##           High    Low    Med  <NA>
##  0           0 72474      0      0
##  1           0 36277      0      0
##  2           0      0 26066      0
##  3           0      0 19012      0
##  4           0      0 14908      0
##  5           0      0 10306      0
##  6      5905      0      0      0
##  7      3382      0      0      0
##  8      1878      0      0      0
##  9      2979      0      0      0
## <NA>         0      0      0 50910
```

```
cat("## Table for pmsu_yn x pmsu_lmh")
```

```
## ## Table for pmsu_yn x pmsu_lmh
```

```
table(dat$pmsu_lmh, dat$pmsu_yn, useNA = "always")
```

```
##
##           0      1  <NA>
##  High           0 14144      0
##  Low  108751      0      0
##  Med   70292      0      0
##  <NA>         0      0 50910
```

Family Support

There are 4 of them. The question starts with: We are interested in how you feel about the following statements. Please show how much you agree or disagree with each one. Please tick one box for each line.

The options to answer go from 1 Very strongly disagree to 7 Very strongly agree. The 4 questions are:

```
family_items <- data.frame(
  item = c(
    "famhelp",
    "famsup",
    "famtalk",
    "famclose"
  ),
  topic = c(
    "Family tries to help",
    "Get emotional help from family",
    "Talk about problems with family",
    "Feel close to family"
  )
)

kable(family_items) #, caption = "Family Support Items")
```

item	topic
famhelp	Family tries to help
famsup	Get emotional help from family
famtalk	Talk about problems with family
famclose	Feel close to family

These variables are then combined to create a categorical predictor. For now, I will keep two options like I did for PMSU but will select only one when modeling. These are:

- a) H/L based on avg of 4 vars. L if [1,5.5) and H if [5.5, 7]
- b) H/M/L based on sum of 4 vars. L if 4-11, M if 12-19, H if 20-28.

```
vars_family_support <- c("famhelp", "famsup", "famtalk", "famdec")

dat <- dat %>%
  mutate(family_support_avg = rowMeans(dat[,vars_family_support]),
         family_support_sum = rowSums(dat[,vars_family_support])
  ) %>%
  mutate(family_support_high = case_when(
    family_support_avg >= 1 & family_support_avg < 5.5 ~ 0,
    family_support_avg >= 5.5 & family_support_avg <= 7 ~ 1,
    TRUE ~ NA)) %>%
  mutate(family_support_hml = case_when(
    family_support_sum >= 4 & family_support_sum <= 11 ~ "Low",
    family_support_sum >= 12 & family_support_sum <= 19 ~ "Med",
    family_support_sum >= 20 & family_support_sum <= 28 ~ "High",
    TRUE ~ NA))
```

Their counts after recoding are:

```

# #table(dat$family_support_high, dat$family_support_hml, useNA = "always")
# cat("## Table for family_support_high")
# table(dat$family_support_high, useNA = "always")

#cat("## Table for family_support_hml")
#table(dat$family_support_hml, useNA = "always")

t <- data.frame(
  table(dat$family_support_high, useNA = "always")
) %>%
  mutate(pct = round(Freq / sum(Freq),3))

kable(t, caption = "family_support_high")

```

Table 3: family__support__high

Var1	Freq	pct
0	65676	0.269
1	156068	0.639
NA	22353	0.092

```

t <- data.frame(
  table(dat$family_support_hml, useNA = "always")
) %>%
  mutate(pct = round(Freq / sum(Freq),3))

kable(t, caption = "family_support_hml")

```

Table 4: family__support__hml

Var1	Freq	pct
High	169660	0.695
Low	23345	0.096
Med	28739	0.118
NA	22353	0.092

Friends (Peer) Support

There are 4 friends support questions. The questions start with: We are interested in how you feel about the following statements. Please show how much you agree or disagree with each one. Please tick one box for each line.

The answers range from 1 Very strongly disagree to 7 Very strongly agree.

```

friend_items <- data.frame(
  item = c(
    "friendhelp",
    "friendcounton",
    "friendshare",

```

```

    "friendtalk"
  ),
  topic = c(
    "My friends really try to help me",
    "I can count on my friends when things go wrong",
    "I have friends with whom I can share my joys and sorrows",
    "I can talk about my problems with my friends"
  )
)
kable(friend_items)

```

item	topic
friendhelp	My friends really try to help me
friendcounton	I can count on my friends when things go wrong
friendshare	I have friends with whom I can share my joys and sorrows
friendtalk	I can talk about my problems with my friends

These variables are then combined to create a categorical predictor. For now, I will keep two options like I did for family but will select only one when modeling. These are:

- a) H/L based on avg of 4 vars. L if [1,5.5) and H if [5.5, 7]
- b) H/M/L based on sum of 4 vars. L if 4-11, M if 12-19, H if 20-28.

```

vars_friends_support <- c("friendhelp", "friendcounton", "friendshare", "friendtalk")
# recode friends/peer support as yn and hml
dat <- dat %>%
  mutate(friends_support_avg = rowMeans(dat[,vars_friends_support]),
         friends_support_sum = rowSums(dat[,vars_friends_support])
  ) %>%
  mutate(friends_support_high = case_when(
    friends_support_avg >= 1 & friends_support_avg < 5.5 ~ 0,
    friends_support_avg >= 5.5 & friends_support_avg <= 7 ~ 1,
    TRUE ~ NA)) %>%
  mutate(friends_support_hml = case_when(
    friends_support_sum >= 4 & friends_support_sum <= 11 ~ "Low",
    friends_support_sum >= 12 & friends_support_sum <= 19 ~ "Med",
    friends_support_sum >= 20 & friends_support_sum <= 28 ~ "High",
    TRUE ~ NA))

```

Their counts after recoding are:

```

t <- data.frame(
  table(dat$friends_support_high, useNA = "always")
) %>%
  mutate(pct = round(Freq / sum(Freq),3))
kable(t, caption = "friends_support_high")

```

Table 6: friends_support_high

Var1	Freq	pct
0	95821	0.393
1	135380	0.555
NA	12896	0.053

```
t <- data.frame(
  table(dat$friends_support_hml, useNA = "always")
) %>%
mutate(pct = round(Freq / sum(Freq),3))

kable(t, caption = "friends_support_hml")
```

Table 7: friends_support_hml

Var1	Freq	pct
High	155201	0.636
Low	29537	0.121
Med	46463	0.190
NA	12896	0.053

```
# table(dat$friends_support_high, dat$friends_support_hml, useNA = "always")
# table(dat$friends_support_high, useNA = "always")
# table(dat$friends_support_hml, useNA = "always")
```

School support

This metric is a combination of multiple questions from 2 categories: teacher acceptance and classmates acceptance. The answers to these questions go from 1 (strong agree) to 5 (strong disagree).

The questions are:

```
school_items <- data.frame(
  category = c("teacher", "teacher", "teacher",
               "classmates", "classmates", "classmates"),
  item = c("teacheraccept",
            "teachercare",
            "teachertrust",
            "studtogether",
            "studhelpful",
            "studaccept"),
  topic = c("Teacher accepts me",
            "Teacher cares about me",
            "Feel trust in teacher",
            "Students enjoy being together",
            "Students kind and helpful",
            "Students accept me")
)

kable(school_items)
```

category	item	topic
teacher	teacheraccept	Teacher accepts me
teacher	teachercare	Teacher cares about me
teacher	teachertrust	Feel trust in teacher
classmates	studtogether	Students enjoy being together
classmates	studhelpful	Students kind and helpful
classmates	studaccept	Students accept me

The scales of these variables is reversed so 1 is negative and 5 is the most positive. Then an arithmetic average is calculated for teacher and students. The based on this average a high/low support is calculated for each with 4 being the threshold.

```
# recodes from 1-5 to 0-4 (06-italy)
recode_school_vars <- function(old_values){

  new_values <- case_when(
    old_values == 5 ~ 1,
    old_values == 4 ~ 2,
    old_values == 3 ~ 3,
    old_values == 2 ~ 4,
    old_values == 1 ~ 5,
    TRUE ~ NA
  )

  return(new_values)
}

dat <- dat %>%
  # recode they can be added
  mutate(teacheraccept_r = recode_school_vars(teacheraccept),
         teachercare_r = recode_school_vars(teachercare),
         teachertrust_r = recode_school_vars(teachertrust),
         studtogether_r = recode_school_vars(studtogether),
         studhelpful_r = recode_school_vars(studhelpful),
         studaccept_r = recode_school_vars(studaccept)
  ) %>%
  # derive avg by teacher/student
  mutate(teacher_support_avg = (teacheraccept_r
                               + teachercare_r
                               + teachertrust_r) / 3,
         student_support_avg = (studtogether_r
                               + studhelpful_r
                               + studaccept_r) / 3,
         # just for fun combine and try
         school_support_avg = (teacheraccept_r
                               + teachercare_r
                               + teachertrust_r
                               + studtogether_r
                               + studhelpful_r
                               + studaccept_r) / 6,
  ) %>%
  # derive support yn based on hbsc
```

```
mutate(
  teacher_support_high = case_when(
    teacher_support_avg >= 4 ~ 1,
    teacher_support_avg < 4 ~ 0,
    TRUE ~ NA),
  student_support_high = case_when(
    student_support_avg >= 4 ~ 1,
    student_support_avg < 4 ~ 0,
    TRUE ~ NA),
  school_support_high = case_when(
    school_support_avg >= 4 ~ 1,
    school_support_avg < 4 ~ 0,
    TRUE ~ NA)
)
```

The recoding of each var is:

```
vars_school <- c("teacheraccept",
  "teachercare",
  "teachertrust",
  "studtogether",
  "studhelpful",
  "studaccept")

vars_school_r <- paste0(vars_school,
  "_r")

for (i in seq_along(vars_school)) {
  cat("\n\n###", vars_school[i], "x", vars_school_r[i], "\n")
  print(table(dat[[vars_school[i]]], dat[[vars_school_r[i]]], useNA = "always"))
}
```

```
##
##
## ### teacheraccept x teacheraccept_r
##
##      1      2      3      4      5 <NA>
## 1      0      0      0      0 90101  0
## 2      0      0      0 88287      0      0
## 3      0      0 33135      0      0      0
## 4      0 9562      0      0      0      0
## 5     6248      0      0      0      0      0
## <NA>      0      0      0      0      0 16764
##
##
## ### teachercare x teachercare_r
##
##      1      2      3      4      5 <NA>
## 1      0      0      0      0 58512  0
## 2      0      0      0 81902      0      0
## 3      0      0 56100      0      0      0
## 4      0 18645      0      0      0      0
## 5    10598      0      0      0      0      0
```



```
## <NA>      0      0      0      0      0 18340
##
##
## ### teachertrust x teachertrust_r
##
##          1      2      3      4      5 <NA>
## 1          0      0      0      0 65528      0
## 2          0      0      0 72597      0      0
## 3          0      0 53595      0      0      0
## 4          0 20051      0      0      0      0
## 5      14487      0      0      0      0      0
## <NA>      0      0      0      0      0 17839
##
##
## ### studtogether x studtogether_r
##
##          1      2      3      4      5 <NA>
## 1          0      0      0      0 64685      0
## 2          0      0      0 91183      0      0
## 3          0      0 52410      0      0      0
## 4          0 12772      0      0      0      0
## 5      6696      0      0      0      0      0
## <NA>      0      0      0      0      0 16351
##
##
## ### studhelpful x studhelpful_r
##
##          1      2      3      4      5 <NA>
## 1          0      0      0      0 57252      0
## 2          0      0      0 90713      0      0
## 3          0      0 50364      0      0      0
## 4          0 19809      0      0      0      0
## 5      9213      0      0      0      0      0
## <NA>      0      0      0      0      0 16746
##
##
## ### studaccept x studaccept_r
##
##          1      2      3      4      5 <NA>
## 1          0      0      0      0 79297      0
## 2          0      0      0 87972      0      0
## 3          0      0 37998      0      0      0
## 4          0 12786      0      0      0      0
## 5      8600      0      0      0      0      0
## <NA>      0      0      0      0      0 17444
```

And the actual recoded predictors look like:

```
t <- data.frame(table(dat$teacher_support_high, useNA = "always")) %>%
  mutate(pct = round(Freq / sum(Freq),3))
kable(t, caption = "teacher_support_high")
```

Table 9: teacher_support_high

Var1	Freq	pct
0	100287	0.411
1	124049	0.508
NA	19761	0.081

```
t <- data.frame(table(dat$student_support_high, useNA = "always")) %>%
  mutate(pct = round(Freq / sum(Freq),3))
kable(t, caption = "student_support_high")
```

Table 10: student_support_high

Var1	Freq	pct
0	96707	0.396
1	128592	0.527
NA	18798	0.077

```
t <- data.frame(table(dat$school_support_high, useNA = "always")) %>%
  mutate(pct = round(Freq / sum(Freq),3))
kable(t, caption = "school_support_high")
```

Table 11: school_support_high

Var1	Freq	pct
0	111725	0.458
1	109746	0.450
NA	22626	0.093

IRRELFAS_LMH: Relative family affluence categorical

Derived from IRFAS. So check list of fas-variables in the dataset. It is a cut based on percentiles so I recode to friendly. 1 Lowest 20 pct

2 Medium 60 pct

3 Highest 20 pct

```
dat <- dat %>%
  mutate(IRRELFAS_LMH_r =
    case_when(IRRELFAS_LMH == 1 ~ "Low20",
              IRRELFAS_LMH == 2 ~ "Med60",
              IRRELFAS_LMH == 3 ~ "High20",
              TRUE ~ NA)
  )

t <- table(dat$IRRELFAS_LMH,
  dat$IRRELFAS_LMH_r,
  useNA = "always")
```

```
kable(t,
      caption = "IRRELFAS_LMH x IRRELFAS_LMH_r")
```

Table 12: IRRELFAS_LMH x IRRELFAS_LMH_r

	High20	Low20	Med60	NA
1	0	45818	0	0
2	0	0	142148	0
3	43532	0	0	0
NA	0	0	0	12599

talkfather and talkmother

Two questions measure parent communication. The pre-question is: How easy is it for you to talk to the following persons about things that really bother you? Please tick one box for each line

Answers go from 1 Very easy to 5 Don't have or see.

These are dichotomized for each parent as Yes if [1,2] and No if [3-5].

```
dat <- dat %>%
  mutate(talkfatherYes =
    case_when(talkfather %in% c(1,2) ~ 1,
              talkfather %in% c(3,4,5) ~ 0,
              TRUE ~ NA),
    talkmotherYes =
    case_when(talkmother %in% c(1,2) ~ 1,
              talkmother %in% c(3,4,5) ~ 0,
              TRUE ~ NA)
  )
```

```
cat("## Table for talkfatherYes x talkfather")
```

```
## ## Table for talkfatherYes x talkfather
```

```
table(dat$talkfather, dat$talkfatherYes, useNA = "always")
```

```
##
##      0      1 <NA>
##  1      0 73777      0
##  2      0 78346      0
##  3  40289      0      0
##  4  19010      0      0
##  5  15184      0      0
## <NA>      0      0 17491
```

```
cat("## Table for talkmotherYes x talkmother")
```

```
## ## Table for talkmotherYes x talkmother
```

```
table(dat$talkmother, dat$talkmotherYes, useNA = "always")
```

```
##
##           0           1      <NA>
##  1           0 115899           0
##  2           0  74913           0
##  3        24042           0           0
##  4          9276           0           0
##  5          4446           0           0
## <NA>           0           0 15521
```

I was going to combine to a “number of parents one can talk” but I’ll leave for now and will try as interactions later.

bullying victim

There are 2 variables that measure if kid has been bullied. a) beenbullied: Been bullied past months and b) cbeenbullied: Been cyber bullied. Answers range from 1 Haven’t to 5 Several times a week.

These are dichotomized as Yes if [2-5] and No if [1].

```
dat <- dat %>%
  mutate(beenbulliedYes =
    case_when(beenbullied == 1 ~ 0,
              between(beenbullied, 2, 5) ~ 1,
              TRUE ~ NA),
    cbeenbulliedYes =
    case_when(cbeenbullied == 1 ~ 0,
              between(cbeenbullied, 2, 5) ~ 1,
              TRUE ~ NA)
  )

cat("## Table for beenbullied x beenbulliedYes")
```

```
## ## Table for beenbullied x beenbulliedYes
```

```
table(dat$beenbullied, dat$beenbulliedYes, useNA = "always")
```

```
##
##           0           1      <NA>
##  1    166254           0           0
##  2           0  36624           0
##  3           0  9754           0
##  4           0  6179           0
##  5           0  8324           0
## <NA>           0           0 16962
```

```
cat("## Table for cbeenbullied x cbeenbulliedYes")
```

```
## ## Table for cbeenbullied x cbeenbulliedYes
```

```
table(dat$cbeenbullied, dat$cbeenbulliedYes, useNA = "always")
```

```
##
##           0           1    <NA>
##  1    198900           0         0
##  2           0    20426         0
##  3           0     4625         0
##  4           0     2274         0
##  5           0     2788         0
##  <NA>           0         0    15084
```

age and sex(gender)

There are 3 age groups and 2 genders. I recode to friendly/actual value and they look like this.

```
dat <- dat %>%
  mutate(
    age_r = case_match(agecat,
                       1 ~ 11,
                       2 ~ 13,
                       3 ~ 15,
                       .default = NA
    ),
    sex_r = case_match(sex,
                      1 ~ "Boy",
                      2 ~ "Girl",
                      .default = NA)
  )

cat("## Table for agecat x age_r")
```

```
## ## Table for agecat x age_r
```

```
table(dat$agecat, dat$age_r, useNA = "always")
```

```
##
##           11      13      15    <NA>
##  1     81834         0         0         0
##  2         0    83749         0         0
##  3         0         0    76998         0
##  <NA>         0         0         0    1516
```

```
cat("## Table for sex x sex_r")
```

```
## ## Table for sex x sex_r
```

```
table(dat$sex, dat$sex_r, useNA = "always")
```

```
##
##      Boy  Girl  <NA>
##  1  120350    0    0
##  2      0 123747    0
##  <NA>    0    0    0
```

```
cat("## Table for age_r x sex_r")
```

```
## ## Table for age_r x sex_r
```

```
table(dat$age_r, dat$sex_r, useNA = "always")
```

```
##
##      Boy  Girl  <NA>
##  11  40476 41358    0
##  13  41612 42137    0
##  15  37394 39604    0
##  <NA>   868   648    0
```

Continuos Online Communications

This category measures how often young person communicate online with 4 types of people in their life. Preamble of question is: How often do you have ONLINE contact with the following people? Answers range from 1 (NA or don't know) to 6 (almost all of the time).

Variable	Description
emconlfreq1	Online contact with close friends
emconlfreq2	Online contact with larger friend group
emconlfreq3	Online contact with online friends
emconlfreq4	Online contact with other

I combine this into one dichotomized and adopt the HBSC definition of yes if any of the variables is 6.

```
dat <- dat %>%
  mutate(emconlfreqYes = case_when(
    (emconlfreq1 == 6 | emconlfreq2 == 6 | emconlfreq3 == 6 | emconlfreq4 == 6) ~ 1,
    (!is.na(emconlfreq1) | !is.na(emconlfreq2) | !is.na(emconlfreq3) | !is.na(emconlfreq4)) ~ 0,
    TRUE ~ NA
  ))
#still need to do something with 99

table(dat$emconlfreqYes, useNA = "always")
```

```
##
##      0      1  <NA>
## 143585  74898 25614
```

Prefer Online life

I haven't found a paper that include this but I want to. It asks questions about preference of sharing important stuff online vs real-life. The three variables are:

Variable	Description
emconlpref1	Secrets, more easily online
emconlpref2	Feelings, more easily online
emconlpref3	Concerns, more easily online

The answers range from 1 (Strong disagree) to 5 (Strong agree). The counts are:

```
online_life <- vars <- c("emconlpref1", "emconlpref2", "emconlpref3")

for (i in seq_along(online_life)){
  #print(online_life[i])
  #print(table(dat[online_life[i]], useNA = "ifany"))

  print(
    dat %>%
      group_by(.data[[ online_life[i] ]]) %>%
      tally() %>%
      mutate(pct = round(n/sum(n),3))
  )
}
```

```
## # A tibble: 7 x 3
##   emconlpref1     n  pct
##   <dbl> <int> <dbl>
## 1         1 79803 0.327
## 2         2 43850 0.18
## 3         3 38481 0.158
## 4         4 25303 0.104
## 5         5 15588 0.064
## 6        99 11259 0.046
## 7        NA 29813 0.122
## # A tibble: 7 x 3
##   emconlpref2     n  pct
##   <dbl> <int> <dbl>
## 1         1 73132 0.3
## 2         2 44817 0.184
## 3         3 38415 0.157
## 4         4 29753 0.122
## 5         5 16135 0.066
## 6        99 11259 0.046
## 7        NA 30586 0.125
## # A tibble: 7 x 3
##   emconlpref3     n  pct
##   <dbl> <int> <dbl>
## 1         1 76316 0.313
## 2         2 46124 0.189
## 3         3 39569 0.162
```

```
## 4          4 25442 0.104
## 5          5 14290 0.059
## 6         99 11259 0.046
## 7         NA 31097 0.127
```

For this I copy the school approach which is avg and dichotomize with 4 as threshold:

```
dat <- dat %>%
  mutate(lives_online = (emconlpref1 + emconlpref2 + emconlpref3)/3) %>%
  mutate(lives_onlineYN = ifelse(lives_online >= 4, 1, 0)
  )

table(dat$lives_online %/% 1,
  dat$lives_onlineYN,
  useNA = "always")
```

```
##
##          0      1  <NA>
## 1    76743      0      0
## 2    54007      0      0
## 3    42509      0      0
## 4         0 18411      0
## 5         0  9262      0
## 99         0 11259      0
## <NA>         0      0 31906
```

VOY!!!!!!!!!!!!!! Dataset for Modeling

Only the relevant and prepared variables are exported to a csv so I can start the modeling process with those. File goes to “data/processed/” and includes date.